

Predicting Jeopardy! Winners from Contestant-Board Semantics Analysis

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Occupation-board similarity produced only a small improvement
in winner prediction

Motivation - why might occupation matter?

Occupations may reflect knowledge developed through education and work.

Each Jeopardy! board contains a different mix of subjects.

We tested whether matching a contestant's occupation to the board provides useful information before gameplay begins.

Research Question

Does the match between each contestant's occupation and the clues on a *Jeopardy!* board improve prediction of the game's winner?

Modeling Approach

- Used a multinomial logistic regression model to estimate the probability that each contestant wins a game
- Used Sentence-BERT to compare the meaning of the contestant's occupation and the SOC descriptions with the clues and categories
- Compared its predictions with equal one-in-three probabilities and a simpler model that does not use occupation or clue text

Data Acquisition

Archived SQL tables and clue text were joined into one analysis dataset

Acquisition and integration

1 Cluebase PostgreSQL archive

Downloaded the archived .sql dump and parsed its COPY statements to rebuild game, contestant, and clue tables as CSV files.

2 Wolle / J! Archive clue file

Joined clue text, categories, rounds, and values to Cluebase games by air date. Wolle covered 6,293 games; Cluebase filled 15 unmatched games.

3 Contestant occupation profiles

Extracted occupation phrases from contestant introduction text and removed location wording before semantic comparison.

4 O*NET SOC and NCES CIP-SOC

Added optional occupation descriptions and education-program links, while recording the confidence of each mapping.

11,646

contestant-game records

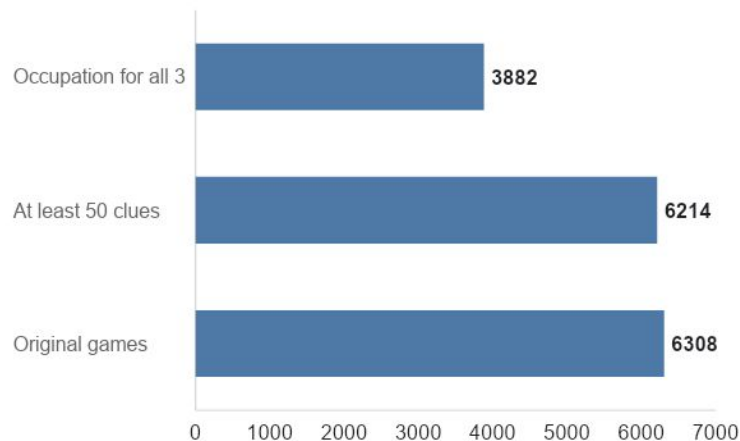
225,388

distinct clues

1984–2019

broadcast dates

Games retained during preprocessing



94 games lacked enough clues; 2,332 lacked an occupation for at least one contestant.

Format: CSV text data. Licensing and Ethics: Public broadcast information was used for academic analysis; third-party ownership remains with the original sources.

Final analysis dataset

Each contestant-game record links occupation text to the clue board and recorded winner

What one record represents

One contestant in one game

Every game contributes three contestant records linked to the same clue board.

Condensed data dictionary

Target	Recorded winner
Main text	Occupation, category, and clue text
Context	Air date, round, clue value, and player position
Excluded	Scores, wagers, responses, and other outcome information

Format and size

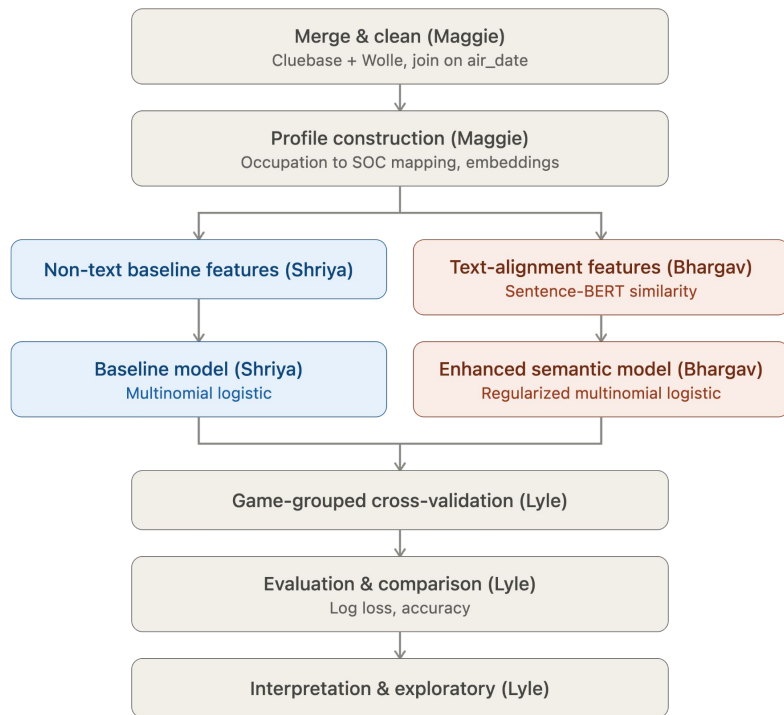
CSV text data

3,882 games 11,646 contestant-game records
225,388 distinct clues Broadcast dates: 1984 to 2019

Main data challenges

- Some boards have missing clues or multimedia content
- Contestant introductions use inconsistent job wording
- Occupation gives only a rough knowledge profile

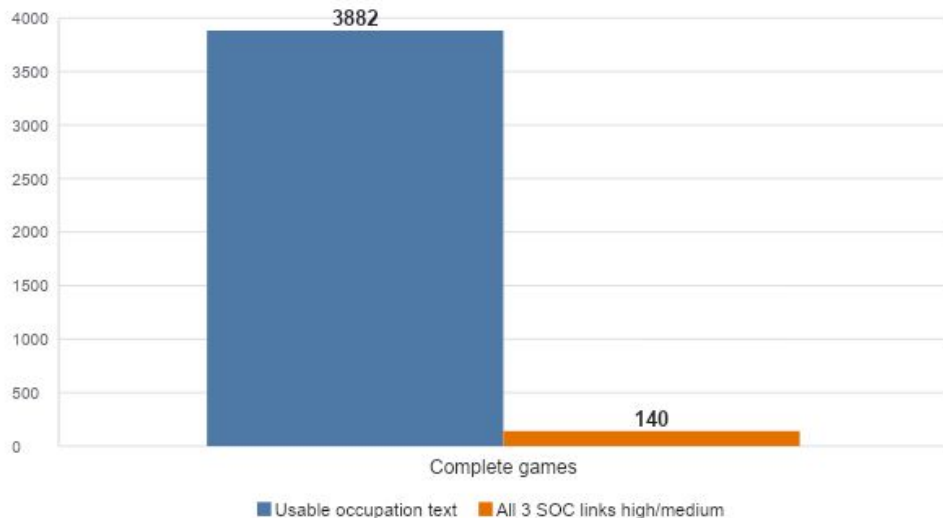
Analysis Plan



- Comparison of two logistic regression models, trained on identical data
 - Both trained on 80% of the games and tested on 20%
- **Baseline:**
 - Only uses standing position to predict winner
- **Semantic:**
 - Uses MiniLM vector embedding model to factor in the occupation's similarity to clues on the board
- **Metrics:** log loss (penalizes confident wrong predictions) and top-pick accuracy
- Compare baseline model, semantic model, random guessing (1/3)
- **Goal:** determine not just whether the main model wins, but by how much, and whether the margin is meaningful vs. noise

Tricky Analysis Decision: Occupation mapping involved a sample-size tradeoff

Strict SOC criteria would leave only 140 complete games



140 games = 3.6% of the final sample

Decision

Use the original occupation phrase in the primary model

- Keep all 3,882 complete games
- Record SOC uncertainty
- Reserve SOC/CIP profiles for exploratory checks

3,786 of 11,646 contestant records had high- or medium-confidence SOC mappings.

Impact: the decision preserved statistical power but allowed mapping uncertainty to weaken the semantic signal.

Bias and Uncertainty: Where our results could mislead us

Several biases were corrected for; others remain and are named openly

Biases considered

Corrected for

Peeking at the answer: scores and wagers were kept out of the model.

Train/test overlap: a game never appears in both.

Starting-position edge: both models account for it.

Not corrected for (limitations)

Missing games: only 3,882 of 6,308 had usable data (61.5%).

Job titles are rough: they miss hobbies and education, and matching them to standard job categories was imperfect.

Left out: buzzer speed, clue choice, Daily Doubles, wagering.

Clue gaps: some clues were missing or multimedia.

How we quantified uncertainty

- Tested on 776 unseen games (20% of the data)
- 95% bootstrap intervals on log loss for each model
- Intervals overlap heavily, so we also checked the paired difference

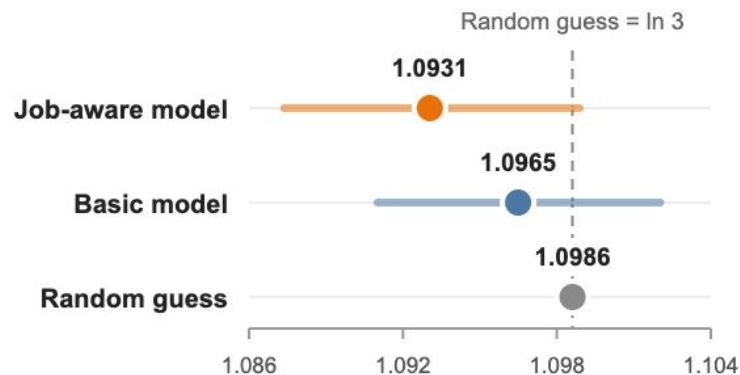
Improvement in prediction error

0.0034

Lower error is better. The plausible range is 0.0002 to 0.0068, barely above zero, so the gain is real but tiny.

Results and Conclusions: Knowing a contestant's job gave only a tiny boost

Does matching a contestant's job to the board's topics help predict who wins? Tested on 776 unseen games.



	Prediction error (lower = better)	Picked the right winner
Job-aware model	1.0930	35.6%
Basic model	1.0965	35.2%
Random guess	1.0986	33.3%

Hypothesis

Matching a contestant's job to the board's topics predicts winners better than a model that ignores jobs.

Result

Supported, barely. Error dropped by 0.0034 (range 0.0002 to 0.0068, just above zero).

Bottom Line

A real but very small edge. Both models are only a little better than guessing.

Next Steps

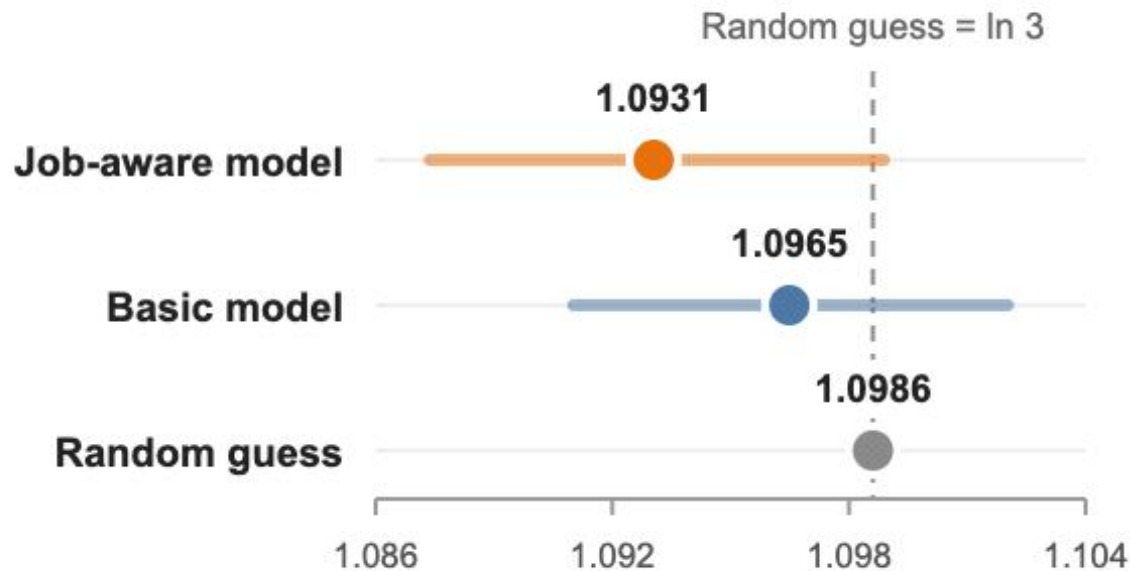
- Fine-tune semantic model to achieve lower log-loss
- Explore comparison to UVA majors
 - Link UVA IRB data (required classes, descriptions, etc) to the CIP-SOC career data used for the original comparison

References

- [1] L. Lavin, “Cluebase documentation.” [Online]. Available: <https://cluebase.readthedocs.io/en/latest/>
- [2] J. Wolle, “Jeopardy clue dataset,” GitHub repository. [Online]. Available: https://github.com/jwolle1/jeopardy_clue_dataset
- [3] National Center for *ONET Development*, “ONET 31.0 occupation data.” [Online]. Available: https://www.onetcenter.org/dl_files/database/db_31_0_json/occupation_data.json
- [4] National Center for Education Statistics, “2020 CIP/SOC crosswalk.” [Online]. Available: https://nces.ed.gov/ipeds/cipcode/Files/CIP2020_SOC2018_Crosswalk.xlsx
- [5] National Center for *ONET Development*, “ONET database license.” [Online]. Available: https://www.onetcenter.org/license_db.html

Questions?

Knowing a contestant's job gave a small edge, but the models' uncertainty ranges overlap heavily



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Prediction error on 776 unseen test games (lower = better). Bars are 95% uncertainty ranges; chart axis truncated.